

The Mediating Field

A substrate-independent account of collective attention:
bioelectric gradients, pheromone trails, neural mean-fields,
and social norms are one mechanism

Internal note — IWAI 2026 paper

2026-05-20

Abstract

Three turns of literature review converged on a single picture, and this note records it. The way a collection of agents “listens” to one another need not be a pairwise aggregation rule that we design by hand. Across wildly different substrates — Levin’s bioelectric morphogenetic fields, Deneubourg’s pheromone trails, the mean-field of a neural population, and the shared norms of a scientific community — nature uses *the same motif*: a slow, shared field sits “in the middle,” every agent deposits into it and reads out of it, and *reading the field is what attention to the collective is*. Pairwise opinion pooling is the special, degenerate case in which each agent privately recomputes the field every step. Promoting the field to a first-class, shared, slowly-relaxing object with its own (possibly nonlinear) dynamics (i) makes “trust” a property of how an agent reads the field rather than a decorative reweighting, (ii) resolves the decentralisation problem of a community-level prior (the field *is* the externalised prior, à la stigmergy), and (iii) relocates the bistability we have been chasing to where it belongs: the self-consistency between fast agents and the slow field.

1 The motif: couple through a field, not pairwise

Our current substrate couples agents pairwise. Each agent i relaxes toward the trust-weighted opinions of its neighbours,

$$\dot{\mu}_i = \underbrace{-\alpha(\mu_i - \theta^*)}_{\text{environment pull}} - \underbrace{\beta \sum_j \Gamma_{ij}(\mu_j - \mu_i)}_{\text{pairwise pooling}}, \quad \sum_j \Gamma_{ij} = 1. \quad (1)$$

The second term is $O(N^2)$ messages, and crucially it is linear in the disagreement $(\mu_j - \mu_i)$, which is exactly why the system is structurally monostable (see the companion diagnosis note). But notice what the sum actually is: define agent i ’s perceived field $f_i := \sum_j \Gamma_{ij} \mu_j$. Then the pooling term is just $-\beta(f_i - \mu_i)$: *agent i is already relaxing toward a field*, it has merely recomputed that field privately, from scratch, this instant, from the raw opinions of others.

Remark 1 (Pooling is privatised field-reading). *DeGroot/linear pooling is the limit of field-mediated coupling in which the field carries no state of its own: it is recomputed each step, shared by no one, and equal to the instantaneous weighted mean. Every richer mechanism below is recovered by giving the field (a) its own slow dynamics, (b) a single shared instance, and (c) a nonlinear deposit or read-out.*

The alternative is to make the field a first-class object f with its own (slow) dynamics,

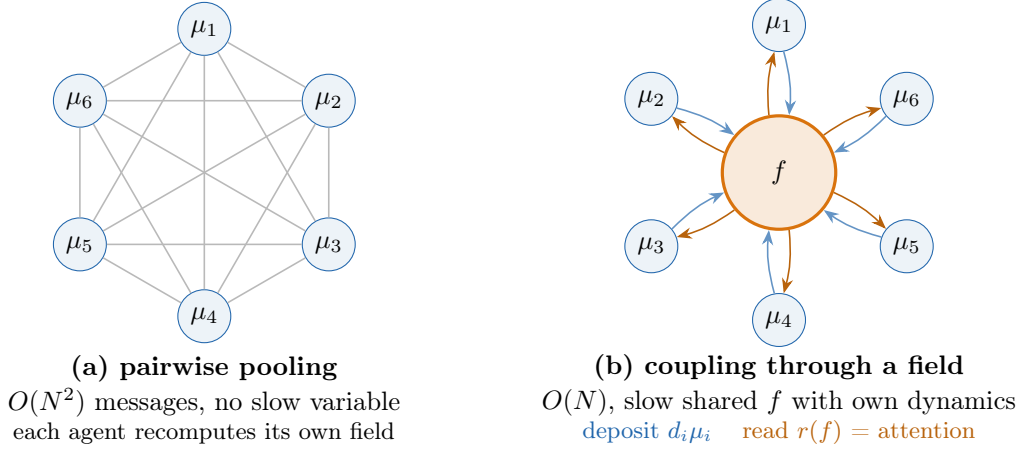


Figure 1: The same coupling, two implementations. **(a)** The complete graph K_6 : agents pool each other’s opinions directly. This is Eq. (1); the “field” exists only as each agent’s private, instantaneous weighted mean. **(b)** A single shared field node f mediates all coupling (Eq. (2)–(3)): agents *deposit* into it (inward, blue) and *read* from it (outward, orange). Reading the field is the act of attending to the collective. The field is slow ($\tau_f \gg 1$) and is the only object that carries collective state between steps.

written-to and read-from by all agents:

$$\dot{\mu}_i = -\alpha(\mu_i - \theta^*) - \beta \pi_i (r(f) - \mu_i), \quad (\text{fast: agents read the field; } \pi_i = \text{how much } i \text{ attends}) \quad (2)$$

$$\tau_f \dot{f} = -f + g\left(\sum_i d_i \mu_i\right), \quad \tau_f \gg 1, \quad (\text{slow: agents deposit } d_i; g = \text{deposit nonlinearity; } \tau_f = \text{evaporation}) \quad (3)$$

Equation (2) is a star coupling through f (Fig. 1b), $O(N)$ rather than $O(N^2)$. The read-out $r(\cdot)$ and the deposit $g(\cdot)$ are where any nonlinearity now lives, and the timescale τ_f is the separation that makes the field a genuine *slow variable* (§4).

2 One motif, four substrates

The claim of this note is that Fig. 1b is substrate-independent. Four literatures describe the *same* star-through-a-field, differing only in what the field is made of and in the shape of the deposit/read-out nonlinearities (Fig. 2, Table 1).

3 Reading the field is attention

In Eq. (2) the gain π_i on the read-out term is precisely the *precision* an agent assigns to the collective field — how much it lets the consensus move its belief. This is the social Hierarchical Gaussian Filter reading of trust [7]: trust is not a row of a mixing matrix but the learned reliability (precision) of a channel. Two limits make the unification concrete.

- **Uniform precision** (π_i equal, $r(f) = f = \text{mean}$): linear averaging, DeGroot, consensus. The honeybee’s proportional response.
- **Peaked precision** (one source dominates): the read-out collapses onto a single voice — “listen to the most knowledgeable.” No separate mechanism; the same field-read at a different precision.

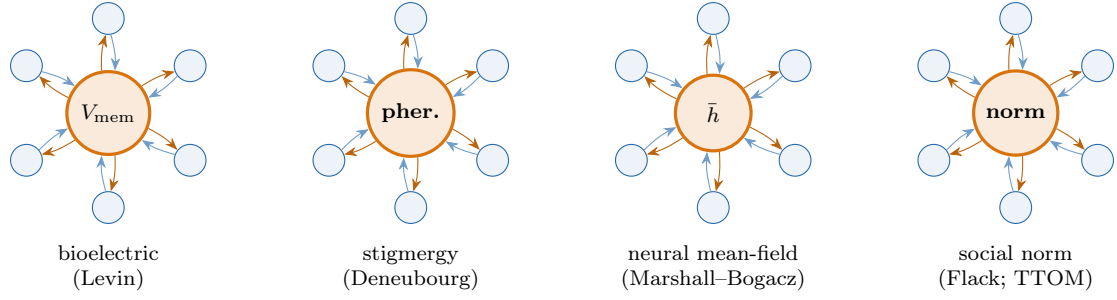


Figure 2: The mediating field across scales. In every case a population of units couples *only* through a slow shared field they both write ($\rightarrow$ in) and sense ($\rightarrow$ out): transmembrane voltage V_{mem} patterning tissue; the pheromone concentration on a trail; the mean activity $\bar{h}$ of a competing neural population; the consensus “power structure”/norm a community references. The topology is identical to Fig. 1b.

The community’s *norm* is then the prior on these precisions — a “regime of expectations” [6] specifying how members should weight the field. And because the field is externalised (Table 1), that prior need not be estimated by any central node: it is the standing state of f itself, read by all. This is the stigmergic resolution of the decentralisation problem — the norm lives in the trail, not in a head.

4 The nonlinearity decides the phase

With the field explicit, the question “linear or nonlinear pooling?” becomes sharp and is answered by a measured natural experiment in the foraging literature (Fig. 3). If the per-agent response to the field is *proportional* (honeybee), the symmetric consensus state is the only attractor. If it is *comparative and amplifying* (ant; Deneubourg’s $q^n/(q^n + (1 - q)^n)$), the symmetric state goes unstable and the field breaks symmetry into one of two locked states — bistability, i.e. paradigm lock-in.

Proposition 1 (Field-mediated bistability). *Let agents relax quickly to the field (α, β fast) and let the field obey $\tau_f \dot{f} = -f + F(f)$ with F the composition of deposit and read-out over the (fast-equilibrated) population. If $F'(f^*) > 1$ at the symmetric fixed point, that point is unstable and, for bounded sigmoidal F , two stable fixed points exist. Bistability is thus a property of the slow field’s self-consistency loop, not of any single agent’s update.*

This is the honest location of the phenomenon. Linear pooling (Fig. 3a, grey) keeps $F' \leq 1$ and cannot fold; a comparative/cross-inhibitory read-out pushes $F' > 1$ and can. It also says *why* an over-informative environment defeats the effect: a large α pull toward θ^* flattens the effective F , and the relevant competition is F' vs 1 once the environment is folded in — a clean, interpretable line rather than a magic coefficient.

5 Consequences for the active-inference rebuild

1. **Add a field, not more edges.** Represent f explicitly as a slow shared state (a stigmergic “trail” over paradigms). In the categorical `pymdp` rebuild this is a shared object agents observe as a modality and deposit into — $O(N)$, and it carries the collective memory between steps.
2. **Put the nonlinearity in the read-out, derived not designed.** Use comparative/cross-inhibitory reading (the optimal-decision motif of [4]) so $F' > 1$ is reachable. Avoid a

	Bioelectric	Stigmergic	Neural	Social
Field f	transmembrane voltage V_{mem}	pheromone conc.	pop. mean activity $\bar{h}$	shared norm / power
Deposit g	ion-channel flux	trail laying	spiking	public claim, citation
Evaporate $1/\tau_f$	gap-junction leak	chemical decay	adaptation	forgetting, turnover
Read = attention	sense V_{mem} gradient	follow gradient	recurrent input	defer to norm
Nonlinearity	voltage-gated channels	comparative trail choice	cross-inhibition	comparative status
Bistable?	pattern memory	symmetry-broken trail	DDM decision	paradigm lock-in
Reference	[1, 2]	[3]	[4]	[5, 6]

Table 1: One motif, four substrates. Rows are the components of Eqs. (2)–(3); columns are the physical realisations. The shared structure is the point: a slow field, written and read, with the listening nonlinearity living in the deposit/read-out.

hand-tuned tanh on the coupling; derive the inhibition from the decision the agent is actually making.

3. **Trust = read-out precision π_i , with the norm as its prior.** This makes trust load-bearing (it sets F' , hence the phase) rather than decorative, and frames the community norm as the standing field state.
4. **Test the fold, not the trajectory.** The pivotal experiment is whether $F' > 1$ is attainable against the environment's α — measure the self-consistency curve (Fig. 3b) directly rather than hunting for hysteresis in time series.

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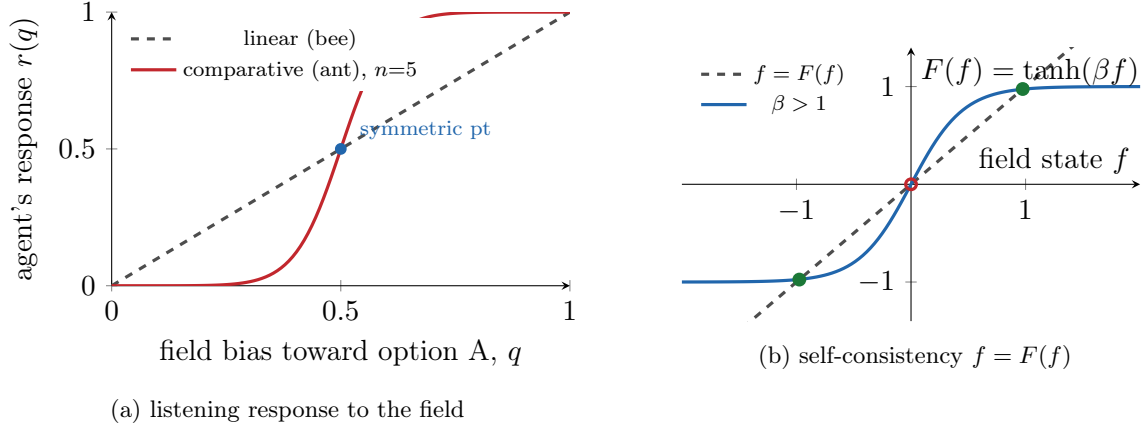


Figure 3: Where bistability comes from. **(a)** The per-agent read-out $r(q)$. Bee = linear/proportional (grey dashed) $\Rightarrow$ slope 1 at the symmetric point $q = \frac{1}{2}$, marginal. Ant = comparative amplification (red) $\Rightarrow$ slope > 1 at $q = \frac{1}{2}$, so the symmetric state is unstable. The *degree of nonlinearity of the listening response* is the bifurcation parameter [3]. **(b)** The same statement as a mean-field self-consistency condition $f = F(f)$ (Flack’s circular causality: the field is computed from the population it then steers). When the slope β of the deposit-read-out loop exceeds 1, the fixed point $f = 0$ (open red, unstable) loses stability and two stable fixed points appear (filled green) — the fold that makes a paradigm self-sustaining [5, 8].

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